



Illuminating
ENGINEERING SOCIETY

TECHNICAL MEMORANDUM:
RAY FILE FORMAT FOR DESCRIPTION
OF THE EMISSION PROPERTIES
OF LIGHT SOURCES
AN AMERICAN NATIONAL STANDARD



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Publication of this Technical
Memorandum has been approved
by IES. Suggestions for revisions
should be directed to IES.

Prepared by:
The Subcommittee on Ray File Format
The IES Computer Committee



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Approved by the IES Standards Committee November 27, 2024, as a Transaction of the Illuminating Engineering Society.

Approved February 25, 2025, as an American National Standard

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Published by the Illuminating Engineering Society, 85 Broad Street, 17th Floor, New York, New York 10004.

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ISBN 978-0-87995-495-6

Printed in the United States of America.

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1.0 Introduction and Scope

1.1 Introduction

In the past few decades, the optical design of illumination systems (non-imaging optics) has benefited greatly from the advances in computer hardware and software. Many commercially available ray-tracing optical design and simulation software programs have been developed to support a wide variety of optical design tasks. All of these software packages can use ray files as source models. Ray files are typically generated by light source manufacturers using either an optical simulation or physical measurements using near field goniometers. These ray files are then put in a specific format for each optical design program, which allows optical engineers to integrate the light source characteristics into their optical system design simulations. Ray files describe light sources by a large number of rays with individual start location, direction, flux, and optional spectral and/or polarization data.

Ray files are mainly intended for optical engineers, who are designing optical systems using light sources. Ray files are a characterization of the near field characteristics of a light source. Users may also be familiar with files formatted according to *ANSI/IES LM-63-19(R25), Approved Method: IES, Standard File Format for the Electronic Transfer of Photometric Data and Related Information*. These are typically called “IES files” and are far-field characterizations of a lamp or luminaire. IES files are used by lighting engineers and designers to calculate predicted light levels. Although these far-field files are useful for calculating lighting application layouts, they are not adequate when designing optics placed in close proximity to the light source.

One complication in providing ray files for light source characterization is that each ray file can be very large, containing millions of rays and using more than 250 MB of data per file. Currently, a manufacturer needs to provide ray files for all of its light sources, in up to 16 unique file formats since none of the commercially available ray tracing software uses the same file format. This situation leads to a large amount of data that has to be created, managed, stored, and distributed to ray file users.

Because all ray files store essentially the same data, the industry has come together to create a single standard ray file format that all light source manufacturers can distribute and all optical design software can use for simulations. The format has been created by a group that includes LED manufacturers, optical simulation software vendors, near-field goniometer vendors, and academia. The new format is meant to allow simple conversion of all existing ray files into the new format, and it will be generally backward compatible in future revisions.

This document defines the IES standardized ray file format to describe the emission properties of light sources that can be used in all commercially available design, analysis, and metrology software.

(Note: IES TM-25-13 Addendum A has been incorporated into this document. Refer to **Section 3.21** for information on the polarization ellipse definition that was added via the Addendum.)

1.2 Scope

This document provides recommendations for a standard ray file format to describe the emission properties of light sources. The ray file format contains information necessary to interface between ray tracing or other optical design, simulation, analysis and metrology software used in lighting applications.

2.0 Normative References

2.1 ANSI/IES LS-1-22

Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2022. Online: www.ies.org/standards/definitions/. (Accessed 2025 Mar 3).

2.2 ISO 8601

Date and Time Format, Parts 1 and 2. Geneva: International Organization for Standardization, 2019.

2.3 IEEE 754-2008

IEEE Standard for Floating-Point Arithmetic. New Jersey: Institute of Electrical and Electronics Engineers; 2008.

2.4 The Unicode Standard

The Unicode Standard. Mountain View, Calif.: The Unicode Consortium.

2.5 The International System of Units (SI)

The International System of Units (SI), 8th ed. Paris: International Bureau of Weights and Measures (BIPM); 2006.

2.6 ISO/IEC 8859-1:1998

Information Technology – 8-bit single-byte coded graphic character sets – Part 1: Latin alphabet No. 1. Geneva: International Organization for Standardization, 1998.

3.0 Definitions and Nomenclature

3.1 ASCII

An encoding scheme for Latin characters, according to ISO/IEC 8859-1 (see **Section 2**). For the purpose of this standard, only the character 0 and characters from 32 up to and including 126, in decimal form, shall be used.

3.2 binary

An encoding of information into certain data types in files creating a sequence of items that can be used immediately as data by computer processors without further conversion; the sequence of binary items is, in general, not a sequence of encoded human-readable characters.

3.3 binary32

A certain binary floating-point encoding format defined in IEEE 754 (see **Section 2**), using four bytes of memory per floating-point number. The binary32 format is commonly used by C and C++ compilers to implement the “float” data type.

3.4 byte

A unit of digital information, consisting of 8 bits.

3.5 denormalized real

As defined in IEEE 754 (see **Section 2**): a number whose absolute value is nonzero but smaller than 1.18×10^{-38} , and thus too small to be encoded within the IEEE 754 binary32 floating-point format at full mantissa resolution.

3.6 direction cosine

Ray directions within the scope of this Standard shall be given as unit vectors in a three-dimensional Cartesian coordinate system (a unit vector is normalized to have a 2-norm, or Euclidean length, of 1). Each coordinate value of a unit vector is mathematically equal to the cosine of the angle between the vector and the corresponding axis. Therefore, the ray direction coordinates are often called *direction cosines*.

3.7 fatal error

When any data in an IES TM-25 ray file shows a condition denoted “fatal error” within this Standard, then the ray file shall be considered invalid.

3.8 light source

For the purpose of this Standard: Any object producing optical radiation (electromagnetic radiation with wavelengths between approximately 100 nm and 1 mm), whether human-made or natural.

3.9 Little Endian

The ordering of individual bytes within larger data items, e.g., binary32 floating point numbers; the least significant byte is stored first.

3.10 lumen

The SI unit of luminous flux.

3.11 luminous flux, Φ_v

As defined in ANSI/IES LS 1-22 (see **Section 2**): radiant flux; the time-rate flow of radiant energy, evaluated in terms of a visual response:

$$\phi_v = K_m \int \phi_{e,\lambda} V(\lambda) d\lambda$$

3.12 NaN (quiet NaN, signaling NaN)

In computing: a numeric data type value representing an undefined or un-representable value, especially in floating-point calculations, per IEEE 754 (see **Section 2**). If $a = 1$, it is a **quiet NaN**; if a is zero and the payload is nonzero, then it is a *signaling NaN*. (Note: NaN stands for *not a number*.)

3.13 normalized real

As defined in IEEE 754 (see **Section 2**): a number whose absolute value is in the range between 1.18×10^{-38} and 3.4×10^{38} , and therefore encodable within the four-byte binary32 format at full mantissa resolution.

3.14 parser

A computer program capable of syntactic analysis of a structured data file, such as a ray file.

3.15 radiant flux

As defined in ANSI/IES LS-1-22: the time rate of flow of radiant energy, expressed preferably in watts.

3.16 ray

An idealized narrow beam of light, with negligible spatial and angular extent. Mathematically, a ray is described as a straight half-line, unambiguously given by its ray start position and its ray direction.

3.17 ray direction

A unit vector in three-dimensional Euclidean space.

3.18 ray start position

A point in three-dimensional Euclidean space.

3.19 ray file

A block of information about light sources, stored in and available from durable storage (e.g., a hard disk).

3.20 signaling NaN

(See **Section 3.12 NaN**.)

3.21 Stokes parameters

Four certain numbers, S_0 , S_1 , S_2 , and S_3 are used to describe the polarization state of electromagnetic radiation along a ray, as defined in standard textbooks on electromagnetic radiation. They form a four-dimensional vector (S_0 , S_1 , S_2 , S_3), called the *Stokes vector*. S_0 is defined as the total radiant flux; S_1 and S_2 describe the amount and orientation of linear polarization; and S_3 describes the amount and orientation of circular polarization. For unpolarized light, $S_1=S_2=S_3=0$, and for fully polarized light $(S_1)^2 + (S_2)^2 + (S_3)^2 = (S_0)^2$. Within this standard, the values of S_1 , S_2 , and S_3 are normalized by division by S_0 , such that $0 \leq (S_1)^2 + (S_2)^2 + (S_3)^2 \leq 1$, and $(S_1)^2 + (S_2)^2 + (S_3)^2$ is equal to the polarization fraction.

A polarized ray requires a geometrical reference frame for the polarization ellipse described by the 4element Stokes vector. Another unit vector, orthogonal to the ray direction unit vector, is required to provide the frame of reference, which is a right-handed coordinate system. The unit vector that is recommended is one that points in the direction of the major axis of the polarization ellipse described by the Stokes vector. This vector is used because it has physical significance. **Table 4-2** includes this information, which is described in **Sections 4.8.1.13** through **4.8.1.19** tristimulus values of a light, X, Y, Z.

As defined in ANSI/IES LS-1-22: the amounts of each of three specific primaries required to match the color of the light. The actual values are obtained by integration of spectral radiant flux using the three color matching functions defined by the International Commission on Illumination (CIE).

4.0 Ray File Format

4.1 File Type and Extension

The file type shall be binary, and shall be written as Little Endian. A discussion of the file type selection is included in **Annex A**.

In order to provide brand recognition for IES TM-25-13 ray files, as well as avoid confusion with legacy ray files, the file extension of “.tm25ray” shall be used for all IES TM-25-13-compatible ray files.

4.2 Overall File Structure

The binary file has been structured into two major sections with several sub-section blocks as shown in **Table 4-1**. The goal is to separate the header information from the ray data to simplify reading of the file and make backward and future compatibility simpler.

Table 4-1. Overall File Structure

Section	Present	Size
Header Section	Always	Variable
File Header	Always	256 bytes
Known Data Flags	Always	32 bytes
Description Header	Always	36,000 bytes
Spectral Tables	Not Always	Variable
Additional Ray Data Column Labels	Not Always	Variable
Additional Text	Not Always	Variable
Ray Data Section	Always	Variable
Ray Data Lines	Always	Variable

4.3 Ray Ordering and Sampling

Although anyone generating a ray file may choose to write the rays in a particular order (e.g., sorted by position or direction cosines), there are no specific requirements for the order in which rays appear in the file. As a result, users need to proceed with caution when attempting to reconstruct the source distribution by reading only parts of the file, as doing so could lead to erroneous results (e.g., inadequate sampling of the spatial and/or angular distribution of the source). It is recommended that all of the rays in the file are read in by the user when reconstructing the source distribution rather than sampling only parts of the file.

4.4 Units in File

The file units shall be millimeters (mm) for dimension and position values, nanometers (nm) for wavelength, watts (W) for radiant flux values, and lumens (lm) for luminous flux values.

4.5 Data Types in File

The file uses the data types identified in **Sections 4.5.1** through **4.5.5**.

4.5.1 ASCII. An ASCII string within the standard ASCII character set. In IES TM-25-13 ray files, ASCII strings appear only in blocks of known size. To express strings shorter than the block size, all unused characters shall be set to ASCII null (a byte with all bits set to zero). A string beginning with the ASCII null character is an empty string.

4.5.2 Int. A 4-byte signed integer of a range -2,147,483,648 to 2,147,483,647. Generally, all integers in the file shall be positive (refer to each field for specific conditions).

4.5.3 Float. A 4-byte binary32 number based on IEEE 754 (see **Section 2**). Generally, valid floats shall be positive and negative normalized reals, zero, and signaling NaNs. Other values (i.e., denormalized, infinity, and quiet NaNs) make the file invalid. Signaling NaNs shall be used to denote “missing” or “not provided.”

4.5.4 String. A UTF-32 text string based on Unicode. In IES-TM-25-13 ray files, strings appear only in blocks of known size. To express strings shorter than the block size, all unused characters shall be set to null (\u0000, a sequence of four bytes with all bits set to zero). A string beginning with the null (\u0000) character is an empty string.

4.5.5 Unsigned Long Int. An 8-byte long or big integer of a range from 0 to 18,446,744,073,709,551,615.

4.6 Section and Block Breakdown

Table 4-2 shows each section of the file, along with the size, data type, and example data, as well as a reference to the appropriate section in this document where the field is described in further detail (including a full description of the field, valid values for the field, and error conditions).

The column “Required in File” indicates which fields are required for the file to be a valid ray file. This simply indicates whether the field is required to be present, not whether it has any particular content. Fields are required to be present so that the first portion of the header stays a fixed size, while particular content requirements are detailed in the following sections.

Table 4-2. Section and Block Breakdown

Document Section	Field Name	Required in File	Size in Bytes	Data Type	Example Data
4.7	Header Section	Yes	variable		
4.7.1	File Header Block	Yes	256		
0	File Type	Yes	4	ASCII	“TM25”
0	File Version	Yes	4	Int	2013
0	Creation Method	Yes	4	Int	0
0	Total Luminous Flux	Yes	4	Float	2300.12
0	Total Radiant Flux	Yes	4	Float	6700.59
0	Number of Rays	Yes	8	Unsigned Long Int	1000000
0	File Creation Date and Time	Yes	28	ASCII	“2013-09-04T08:30:29+01:00”
0	Ray Start Position	Yes	4	Int	0
0	Spectral Data Identifier	Yes	4	Int	1
0	Single Wavelength	Yes	4	Float	555.5
0	Minimum Wavelength	Yes	4	Float	450.0
0	Maximum Wavelength	Yes	4	Float	750.1
0	Number of Spectral Tables	Yes	4	Int	1
0	Number of Additional Ray Data Items Per Ray	Yes	4	Int	0
0	Size of Additional Text Block	Yes	4	Int	0
0	Reserved for Future Use	Yes	168		
4.7.2	Known Data Flags Block	Yes	32		
0	Position Flag	Yes	4	Int	1
0	Direction Flag	Yes	4	Int	1
0	Radiant Flux / Stokes S0 Flag	Yes	4	Int	1
0	Wavelength Flag	Yes	4	Int	0
0	Luminous Flux / Y Tristimulus Flag	Yes	4	Int	0
0	Stokes Flag	Yes	4	Int	0
0	Tristimulus Flag	Yes	4	Int	0
0	Spectrum Index Flag	Yes	4	Int	0

Table 4-2. Section and Block Breakdown (continued)

Document Section	Field Name	Required in File	Size in Bytes	Data Type	Example Data
4.7.3	Description Header Block	Yes	36,000		
0	Name of Light Source	Yes	4,000	String	"1W LED"
0	Manufacturer of Light Source	Yes	4,000	String	"LED Company"
0	Creator of Optical Light Source Model	Yes	4,000	String	"Engineer 1"
0	Creator of the Ray File	Yes	4,000	String	"Measurement Company"
0	Measurement Equipment / Simulation Software	Yes	4,000	String	"Ray Maker 2000"
0	Camera Information	Yes	4,000	String	"Fixed Focus"
0	Light Source Operation Information	Yes	4,000	String	"350mA, 25C"
0	Additional Information	Yes	4,000	String	"sample LED 1 of 12"
0	Data Reference to Light Source Geometry	Yes	4,000	String	"top of dome is 1mm above ray file origin"
4.7.4	Spectral Tables Block	No	variable		
	Number of Pairs for Table 1	No	4	Int	41
	Wavelength #1, Table 1	No	4	Float	350.1
	Relative Weight #1, Table 1	No	4	Float	0.5
	... (data pairs)	See Spectral Tables Block, Section 4.7.4 for formats			
	Wavelength #N, Table 1	No	4	Float	780.2
	Relative Weight #N, Table 1	No	4	Float	0.25
	... (spectral tables)	See Spectral Tables Block, Section 4.7.4 for formats			

	Number of Pairs for Table M,	No	4	Int	41
	Wavelength #1, Table M	No	4	Float	350.3
	Relative Weight #1, Table M	No	4	Float	0.25
	... (data pairs)	See Spectral Tables Block (Section 4.7.4)			
	Wavelength #P, Table M	No	4	Float	780.4
	Relative Weight #P, Table M	No	4	Float	0.8
	... (spectral tables)	See Spectral Tables Block (Section 4.7.4)			
4.7.5	Additional Ray Data Column Labels Block	No	variable		
	Ray Data Column Name #1	No	512	String	"Photosynthetic Photon Flux [micro-moles]"
	... (column names)	No	512 (each)	String	
	Ray Data Column Name #Q	No	512	String	
4.7.6	Additional Text Block	No	variable		
	See Additional Text Block (Section 4.7.6)				
No	as defined in header	String			

Table 4-2. Section and Block Breakdown *(continued)*

Document Section	Field Name	Required in File	Size in Bytes	Data Type	Example Data
4.8	Ray Data Section	Yes	variable		
4.8.1	Ray Data Lines Block	Yes	variable		
0	X Position	Yes	4	Float	0.0
0	Y Position	Yes	4	Float	0.0
0	Z Position	Yes	4	Float	0.0
0	X Direction Cosine	Yes	4	Float	0.0
0	Y Direction Cosine	Yes	4	Float	0.0
0	Z Direction Cosine	Yes	4	Float	1.0
0	Radiant Flux / Stokes S0	No	4	Float	100.0
0	Wavelength	No	4	Float	350.0
0	Y Tristimulus / Luminous Flux	No	4	Float	20.8
0	Stokes S1	No	4	Float	1.0
0	Stokes S2	No	4	Float	0.0
0	Stokes S3	No	4	Float	0.0
4.8.1.13	Polarization Ellipse Major Axis X Direction Cosine	No	4	Float	1.0
4.8.1.14	Polarization Ellipse Major Axis Y Direction Cosine	No	4	Float	0.0
4.8.1.15	Polarization Ellipse Major Axis Z Direction Cosine	No	4	Float	0.0
06	X Tristimulus	No	4	Float	95.047
0	Z Tristimulus	No	4	Float	108.883
0	Spectrum Index	No	4	Int	1
0	Additional User Defined Data	No	4 per addl item	Float	See Additional User Defined Data (Section 4.8.1.19)

4.7 Header Section

The Header Section contains basic information about the light source model that does not vary per ray, as well as information necessary to read and interpret the data in the file. Although the size of the Header Section may vary, the first three blocks of the section shall be fixed and contain the data necessary to compute the size of the remainder of the Header Section.

4.7.1 File Header Block. The file header shall be a fixed size block of data containing all of the following fields, whether filled with data or not. Every effort should be made to provide as much data as possible, since this may be useful to the user of the ray file.

4.7.1.1 FILE TYPE

Description:	Shall contain the following string content, which is unique to this ray file format: "TM25"
Example Data:	"TM25"
Valid Data:	"TM25"
Error Conditions:	A file lacking the data identified in Valid Data above is not a valid TM-25 ray file.
Required in File:	Yes
Size in Bytes:	4
Data Type:	ASCII

4.7.1.2 File Version

Description:	Shall contain the year of the file version, in this case: 2013. <i>Note: The file version is not necessary the same as the year the standard was published.</i>
Example Data:	2013
Valid Data:	2013
Error Conditions:	A file lacking the data identified in Valid Data above is not a valid TM-25-13 or TM-25-20 ray file.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.1.3 Creation Method

Description:	Should contain an identification of whether the file was created via measurement or simulation using the following flags: 0 - Simulation 1 - Measurement
Example Data:	0
Valid Data:	Any integer or zero.
Error Conditions:	Values other than 0 or 1 may be used, but shall be interpreted as "not defined" in the software using the ray file for files of this version.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.1.4 Total Luminous Flux

Description:	Shall contain the Total Luminous Flux in lumens.
Example Data:	2300.12
Units:	Lumens
Valid Data:	Positive normalized reals, zero, or signaling NaNs.
Error Conditions:	If both the Total Luminous Flux and the Total Radiant Flux are zero or signaling NaNs, the file is still valid. However, the software using the ray file should prompt the user to enter a flux value. If both Total Luminous Flux and Total Radiant Flux are provided, the radiant flux shall take precedence, or the software using the ray file should prompt the user to select the appropriate value.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Float

4.7.1.5 Total Radiant Flux

Description:	Shall contain the Total Radiant Flux in radiant watts.
Example Data:	6700.59
Units:	Radiant watts
Valid Data:	Positive normalized reals, zero, or signaling NaNs.
Error Conditions:	If both the Total Luminous Flux and the Total Radiant Flux are zero or signaling NaNs, the file is still valid. However, the software using the ray file should prompt the user to enter a flux value. If both Total Luminous Flux and Total Radiant Flux are provided, the radiant flux shall take precedence, or the software using the ray file should prompt the user to select the appropriate value.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Float

4.7.1.6 Number of Rays

Description:	Shall contain the total number of rays contained in the file.
Example Data:	1000000
Valid Data:	Any positive integer.
Error Conditions:	If the Number of Rays does not match the number of Ray Data Lines (Section 4.8.1), the file is invalid.
Required in File:	Yes
Size in Bytes:	8
Data Type:	Unsigned Long Int

4.7.1.7 File Creation Date and Time

Description:	Should contain the file creation date and time. The date and time shall include the year, month, date, hours, minutes, seconds, and time zone designator in ISO 8601 format, YYYY-MM-DDThh:mm:ssTZD
Example Data:	2013-09-04T08:30:29+01:00
Valid Data:	Any date specified in ISO 8601 (see Section 2) format, or any ASCII text string.
Error Conditions:	None; data are not required to be complete. If the field does not contain a properly formatted ISO 8601 date (YYYY-MM-DDThh:mm:ssTZD), the reading software may display the contents of the field as a text string.
Required in File:	Yes
Size in Bytes:	28
Data Type:	ASCII

4.7.1.8 Ray Start Position

Description:	Should contain a reference to how the initial ray emission points were determined based on the following flags: 0 – Unknown/Undefined (starting geometry is unspecified) 1 – User (all rays emanate from arbitrary points set by a user) 2 – Plane (all rays emanate from the single side of a plane) 3 – Sphere (all rays emanate from a sphere) 4 – Cylinder (all rays emanate from a cylinder) 5 – Cuboid (all rays emanate from the sides of a cube) 6 – Geometry (all rays emanate from the LED geometry itself) 7 – Back Propagated (all rays emanate from points derived by backward raytracing to find valid light emission surfaces) Additional description information may be provided in Additional Information (Section 4.7.3.8).
Example Data:	0
Valid Data:	Any integer or zero.
Error Conditions:	If the Ray Start Position value is undefined (Option = 0) or not one of the above values, the file is still valid. However, the software using the ray file should warn the user about the lack of this information and shall default to Ray Start Position Option = 0.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.1.9 Spectral Data Identifier

Description:	Shall contain a reference to the manner in which spectral data is used in this file: 0 – No Wavelength Information (watts/lumen based file) 1 – Single Wavelength for all rays (wavelength defined in Section 4.7.1.10) 2 – Wavelength defined per ray (in Section 4.8.1.8) 3 – Single Spectrum for all rays (spectrum defined in Section 4.7.4) 4 – Spectrum defined per ray (spectra defined in Section 4.7.4) See Annex B for a more detailed discussion of the Spectral Data options and their relationships to other data in the TM-25 ray file.
Example Data:	1
Valid Data:	An integer listed in the Description above: 0, 1, 2, 3, 4.
Error Conditions:	If the Spectral Data Identifier value is not one of the Valid Data values, the file shall be invalid, since unknown spectral information makes the entire file uncertain.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.1.10 Single Wavelength

Description:	Shall contain a single reference wavelength in nanometers used for each ray if the Spectral Data Identifier (Section 4.7.1.9) is set to Option = 1 or shall be a signaling NaN for all other values of the Spectral Data Identifier (Section 4.7.1.9).
Example Data:	555.5
Units:	Nanometers
Valid Data:	Any positive number or a signaling NaN.
Error Conditions:	If the Spectral Data Identifier (Section 4.7.1.9) is set to Option = 1 and this field does not contain a positive number, the ray file is invalid. Otherwise, there are no error conditions.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Float

4.7.1.11 Minimum Wavelength

Description:	Shall contain the minimum wavelength in nanometers of all ray wavelengths in Section 4.8.1.8 or spectral tables in Section 4.7.4 if Section 4.7.1.9 is set to Option = 2, 3, or 4. If Section 4.7.1.9 is set to Option = 0 or 1, this field shall be set to a signaling NaN.
Example Data:	450.0
Units:	Nanometers
Valid Data:	Any positive number or a signaling NaN.
Error Conditions:	Negative values may indicate an error in the file but shall not be considered fatal. A minimum wavelength that is more than the maximum wavelength may indicate an error in the file but shall not be considered fatal. The data in this field shall not limit the range of wavelengths that may be read from the spectral tables in Section 4.7.4 or ray data in Section 4.8.1.8. If the minimum wavelength in Section 4.7.1.11 does not agree with the spectral tables or ray data, the spectral tables in Section 4.7.4 or ray data in Section 4.8.1.8 take precedence.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Float

4.7.1.12 Maximum Wavelength

Description:	Shall contain the maximum wavelength in nanometers of all ray wavelengths in Section 4.8.1.8 or spectral tables in Section 4.7.4 if Section 4.7.1.9 is set to Option = 2, 3, or 4. If Section 4.7.1.9 is set to Option = 0 or 1, this field shall be set to a signaling NaN.
Example Data:	750.1
Units:	Nanometers
Valid Data:	Any positive number or a signaling NaN.
Error Conditions:	Negative values may indicate an error but shall not be considered fatal. A maximum wavelength that is less than the minimum wavelength may indicate an error in the file but shall not be considered fatal. The data in this field shall not limit the range of wavelengths that may be read from the spectral tables in Section 4.7.4 or ray data in Section 4.8.1.8. If the maximum wavelength in Section 4.7.1.12 doesn't agree with the spectral tables or ray data, the spectral tables in Section 4.7.4 or ray data in Section 4.8.1.8 take precedence.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Float

4.7.1.13 Number of Spectral Tables

Description:	Shall contain the quantity of spectral tables that will be presented in Spectral Tables Block (Section 4.7.4). This field is ignored unless Spectral Data Identifier (Section 4.7.1.9) is set to Option = 3 or 4.
Example Data:	1
Valid Data:	Any positive integer, zero or a signaling NaN.
Error Conditions:	If Spectral Data Identifier (Section 4.7.1.9) is set to Option = 3 or 4 and this field is not greater than or equal to 1, the ray file is invalid. Otherwise, there are no error conditions.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.1.14 Number of Additional Ray Data Items per Ray

Description:	Shall contain the quantity of additional data items that will be presented for each ray in Additional User Defined Data (Section 4.8.1.19) and labeled in Additional Ray Data Column Labels Block (Section 4.7.5). Each additional item adds a column to the ray data table. Each column is 4 bytes.
Example Data:	0
Valid Data:	Any positive integer or zero.
Error Conditions:	Negative values are a fatal error. Incorrectly setting this value will corrupt the reading of the ray data lines due to a size mismatch.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.1.15 Size of Additional Text Block

Description:	Shall contain the size in bytes of the Additional Text Block that will be presented in Section 4.7.6 . This text block will be a UTF-32 string block. The 2013 version of the file does not specify how this block should be handled or what may be contained. It is anticipated that future versions of the file format will use this Additional Text Block to expand on the file format. Parsers written for 2013 files may safely ignore the additional text block by knowing the size of the block, which is defined in this section.
Example Data:	0
Valid Data:	Any positive integer that is a multiple of 32 (for alignment reasons), or zero.
Error Conditions:	Positive integers that are not a multiple of 32 and any negative integers shall be a fatal error.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.1.16 Reserved for Future Use

Description:	This is reserved space in the Fixed Header Block that may be used in future revisions of the file format. 2013 parsers shall simply ignore any data here if the file is newer than 2013.
Required in File:	Yes
Size in Bytes:	168

4.7.2 Known Data Flags Block

The purpose of the Known Data Flags Block is to provide a way in the fixed header for identifying how much data is present for each ray. Each flag may be set to 0 or 1 to indicate the absence or presence of each data element. Some elements are required, some are optional, and some may not co-exist.

For each of the Known Data Flag fields, a value other than 0 or 1 shall cause the file to be invalid, since the length of the ray data line cannot then be determined.

4.7.2.1 Position Flag

Description:	Shall contain a flag indicating if the data is present in the ray data line. 0 – Absent 1 – Present (adds 3 values to the ray data line)
Example Data:	1
Valid Data:	The integer 1.
Error Conditions:	The Position Flag is always required to be set to 1. If it is not, the ray file is invalid.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.2.2 Direction Flag

Description:	Shall contain a flag indicating if the data is present in the ray data line. 0 – Absent 1 – Present (adds 3 values to the ray data line)
Example Data:	1
Valid Data:	The integer 1.
Error Conditions:	The Direction Flag is always required to be set to 1. If it is not, the ray file is invalid.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.2.3 Radiant Flux Flag / Stokes S0 Vector Flag

Description:	Shall contain a flag indicating if the data is present in the ray data line. 0 – Absent 1 – Present (adds 1 value to the ray data line)
Example Data:	1
Valid Data:	The integer 1 or 0.
Error Conditions:	Either Radiant Flux or Luminous Flux shall be present. If both are absent, the file is invalid. If both are present, the Radiant Flux shall be used. If the Stokes Flag (Section 4.7.2.6) is set to Present, the Radiant Flux Flag (Section 4.7.2.3) shall be set to Present. If it is not, the file is invalid. If the Spectral Data Identifier (Section 4.7.1.9) is set to Option = 2 or 4, the Radiant Flux Flag (Section 4.7.2.3) shall be set to Present. (See Annex B for further information about the use of spectral data, radiant flux, and luminous flux.)
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.2.4 Wavelength Flag

Description:	Shall contain a flag indicating whether data is present in the ray data line. 0 – Absent 1 – Present (adds 1 value to the ray data line)
Example Data:	0
Valid Data:	The integer 1 or 0.
Error Conditions:	Required to be Present if Spectral Data Identifier (Section 4.7.1.9) is set to Option = 2. (See Annex B for further information about the use of spectral data, radiant flux, and luminous flux.)
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.2.5 Luminous Flux / Y Tristimulus Flag

Description:	Shall contain a flag indicating if the data is present in the ray data line. 0 – Absent 1 – Present (adds 1 value to the ray data line)
Example Data:	0
Valid Data:	The integer 1 or 0
Error Conditions:	Either Radiant Flux or Luminous Flux is required. If both are missing the file is invalid. If both are present, the Radiant Flux shall be used. If the Tristimulus Flag (Section 4.7.2.7) is set to Present, the Luminous Flux Flag (Section 4.7.2.5) shall be set to Present. If it is not, the file is invalid. If the Spectral Data Identifier (Section 4.7.1.9) is set to Option = 2 or 4, the Radiant Flux Flag (Section 4.7.2.3) shall be set to Present, and the Luminous Flux Flag (Section 4.7.2.5) shall be set to Absent. (See Annex B for further information about the use of spectral data, radiant flux, and luminous flux.)
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.2.6 Stokes Flag

Description:	Shall contain a flag indicating whether data is present in the ray data line. 0 – Absent 1 – Present (adds 6 values to the ray data line)
Example Data:	0
Valid Data:	The integer 1 or 0.
Error Conditions:	If the Stokes Flag (Section 4.7.2.6) is set to Present, the Radiant Flux Flag (Section 4.7.2.3) shall be set to Present. If it is not, the file is invalid.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.2.7 Tristimulus Flag

Description:	Shall contain a flag indicating whether data is present in the ray data line. 0 – Absent 1 – Present (adds 2 values to the ray data line)
Example Data:	0
Valid Data:	The integer 1 or 0.
Error Conditions:	If the Tristimulus Flag (Section 4.7.2.7) is set to Present, the Luminous Flux Flag (Section 4.7.2.5) shall be set to Present. If it is not, the file is invalid. Tristimulus Data is only allowed if the Spectral Data Identifier (Section 4.7.1.9) is set to Option = 0. (See Annex B for further information about the use of spectral data, radiant flux, and luminous flux.)
Required in File:	Yes
Size in Bytes:	4
Data Type:	Int

4.7.2.8 Spectrum Index Flag

Description:	Shall contain a flag indicating if the data is present in the ray data line. 0 – Absent 1 – Present (adds 1 value to the ray data line) This shall be set to Present when Spectral Data Identifier (Section 4.7.1.9) is set to Option = 4; otherwise, this shall be set to Absent.
Example Data:	0
Valid Data:	The integer 1 or 0.
Error Conditions:	If the value for this flag does not correspond to the option selected in Spectral Data Identifier (Section 4.7.1.9) as identified above, the file is invalid. (See Annex B for further information about the use of spectral data, radiant flux, and luminous flux.)
Required in File:	Yes
Size in Bytes:	Int
Data Type:	4

4.7.3 Description Header Block

The Description Header Block shall be a fixed-size block of data containing all of the following fields, whether filled or not. These fields should be filled out as completely as possible to ensure that users of the ray file know as much about the origin of the data as possible. Each field is allowed 4,000 bytes of UTF-32 string text, for a total of 1,000 characters for each field. If additional description is required, users may include it either in the Additional Text Block (see **Section 4.7.6**) or in URLs in these comment fields to direct users to additional sources of data provided online.

4.7.3.1 Name of Light Source

Description:	Should contain the name of the light source that the ray file describes.
Example Data:	"1W LED"
Valid Data:	Any string
Error Conditions:	None
Required in File:	Yes
Size in Bytes:	4,000
Data Type:	String

4.7.3.2 Manufacturer of Light Source

Description:	Should contain the name of the manufacturer of the light source.
Example Data:	"LED Company"
Valid Data:	Any string
Error Conditions:	None
Required in File:	Yes
Size in Bytes:	4,000
Data Type:	String

4.7.3.3 Creator of Optical Light Source Model

Description:	Should contain the name of the creator of the light source model used to generate the ray file. Uses for this field may be to indicate the creator of the CAD geometry used to simulate the ray data or the name of the technician measuring the ray data.
Example Data:	"Engineer 1"
Valid Data:	Any string
Error Conditions:	None
Required in File:	Yes
Size in Bytes:	4,000
Data Type:	String

4.7.3.4 Creator of the Ray File

Description:	Should contain the name of the company creating the ray file. This may be a third-party or in-house test lab, or in-house engineering group.
Example Data:	"Measurement Company"
Valid Data:	Any string
Error Conditions:	None
Required in File:	Yes
Size in Bytes:	4,000
Data Type:	String

4.7.3.5 Measurement Equipment / Simulation Software

Description:	Should contain the name and a description of the measurement equipment or the simulation software used to gather or generate the ray data.
Example Data:	"Ray Maker 2000"
Valid Data:	Any string
Error Conditions:	None
Required in File:	Yes
Size in Bytes:	4,000
Data Type:	String

4.7.3.6 Camera Information

Description:	Should contain the name and a description of the camera used for the measurement equipment.
Example Data:	"Fixed Focus"
Valid Data:	Any string
Error Conditions:	None
Required in File:	Yes
Size in Bytes:	4,000
Data Type:	String

4.7.3.7 Light Source Operation Information

Description:	Should contain any additional information about the light source that may be relevant to a user of the ray data.
Example Data:	"350mA, 25C"
Valid Data:	Any string
Error Conditions:	None
Required in File:	Yes
Size in Bytes:	4,000
Data Type:	String

4.7.3.8 Additional Information

Description:	Should contain any additional information about the ray file that may be relevant.
Example Data:	"sample LED 1 of 12"
Valid Data:	Any string
Error Conditions:	None
Required in File:	Yes
Size in Bytes:	4,000
Data Type:	String

4.7.3.9 Data Reference to Light Source Geometry

Description:	Should contain a description of how the origin point in the ray data aligns with some point on the light source.
Example Data:	"top of dome is 1mm above ray file origin"
Valid Data:	Any string
Error Conditions:	None
Required in File:	Yes
Size in Bytes:	4,000
Data Type:	String

4.7.4 Spectral Tables Block

This block shall contain M number of spectral tables, where M is defined in **Section 4.7.1.13**. Each table shall be given an index from 1 to M , to be used as reference in the Ray Data lines, **Section 4.8.1.18**. It should be noted that the indices are 1 based and not zero based; therefore, the first spectrum table is the column marked "Table 4-1" in **Table 4-3**.

For each table, there is a set of paired data: wavelength and relative weight. The number of pairs for each table (N , P) is identified before the table begins and does not need to be the same for each of the M tables. **Table 4-3** illustrates the concept of the table. The order of presentation of the elements is defined earlier in this section, and the elements are a series of Ints and Floats.

Computer hardware and software may benefit from keeping data aligned in memory. Therefore, the size in bytes of the spectral tables block shall be a multiple of 32. If the cumulative size in bytes of the spectral tables is not a multiple of 32, then the spectral tables shall be appended by the appropriate number of zero integer values (zero padding).

Table 4-3. Spectral Tables Example

Table 4-1		Table 4-2		...	Table #M	
4 Data Pairs		5 Data Pairs		...	P Data Pairs	
450 nm	1.0	455 nm	0.25	...	300 nm	0.1
600 nm	0.25	500 nm	1.2	...	...	...
700 nm	0.01	555 nm	0.38	...	WL_P	Weight_P
800 nm	0.01	600 nm	0.22	...		
		655 nm	0.19	...		

For the examples given in **Table 4-3**, the data sequence in the ray file is shown in **Table 4-4** (where the *I* and *F* suffixes denote Int and Float, respectively, and where the last seven integer zero values are padding values required to achieve alignment).

Table 4-4. Data Sequence Example

Description	Data Type	Data
Data Pairs for Table 1	Int	4
Wavelength 1	Float	450
Power 1	Float	1
Wavelength 2	Float	600
Power 2	Float	0.25
Wavelength 3	Float	700
Power 3	Float	0.01
Wavelength 4	Float	800
Power 4	Float	0.01
Data Pairs for Table 2	Int	5
Wavelength 1	Float	455
Power 1	Float	0.25
Wavelength 2	Float	500
Power 2	Float	1.2
Wavelength 3	Float	555
Power 3	Float	0.38
Wavelength 4	Float	600
Power 4	Float	0.22
Wavelength 5	Float	655
Power 5	Float	0.19
Data Pairs for Table 3	Int	2
Wavelength 1	Float	300
Power 1	Float	0.1
Wavelength 2	Float	355
Power 2	Float	0.5
Padding	Int	0
Padding	Int	0
Padding	Int	0
Padding	Int	0
Padding	Int	0
Padding	Int	0

4.7.4.1 Data Types. Table indices shall be integers. Data Pair numbers shall be integers. All wavelengths and fluxes shall be floats.

4.7.4.2 Error Conditions. Negative or zero integers shall be fatal errors. Negative or zero wavelengths shall be fatal errors. Negative fluxes shall be fatal errors.

4.7.5 Additional Ray Data Column Labels Block

Description:	For the Number of Additional Ray Data Items Per Ray identified in Section 4.7.1.14 , this block shall provide a name and unit for each column of data. The convention shall be the item name followed by the unit in square brackets: "item name [unit]". Each name field shall be up to 512 bytes of UTF-32 string data.
Example Data:	"Photosynthetic Photon Flux [micro-moles]"
Valid Data:	Any string
Error Conditions:	A file lacking a name for each additional column identified in Section 4.7.1.14 is invalid.
Required in File:	No
Size in Bytes:	512 (for each additional Ray Data Column) Total size shall be 512 multiplied by the integer in Section 4.7.1.14 .
Data Type:	String

4.7.6 Additional Text Block

Description:	Shall contain the amount in bytes of string data identified in Section 4.7.1.15 .
Example Data:	none
Valid Data:	Any string
Error Conditions:	The 2013 format does not specify how to handle this data. It is anticipated that future versions of the file will make use of this area to add additional data. For this reason, parsers for ray data can safely ignore this data and still have access to all of the pertinent information in a 2013 formatted ray file. The goal is that a parser written for the 2013 file format will still be able to read the ray data section of future files, ignoring this Additional Text Block section.
Required in File:	No
Size in Bytes:	As defined in header (multiple of 32)
Data Type:	String

4.8 Ray Data Section

The ray data block contains all of the per-ray information.

4.8.1 Ray Data Lines Block. The ray data line block shall contain all information pertinent to each ray. The elements included for each ray are defined in **Sections 4.7.1** through **4.7.2**. All ray data lines shall contain no signaling NaNs. Signaling NaNs in ray data lines shall be fatal errors.

4.8.1.1 X Position

Description:	Shall contain the x coordinate of the ray start position in mm.
Example Data:	0.0
Units:	Millimeters
Valid Data:	Any real number
Error Conditions:	A file lacking this data is not a valid IES TM-25-13 ray file.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Float

4.8.1.2 Y Position

Description:	Shall contain the y coordinate of the ray start position in mm.
Example Data:	0.0
Units:	Millimeters
Valid Data:	Any real number
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Float

4.8.1.3 Z Position

Description:	Shall contain the z coordinate of the ray start position in mm.
Example Data:	0.0
Units:	Millimeters
Valid Data:	Any real number
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Float

4.8.1.4 X Direction Cosine

Description:	Shall contain the x direction cosine of the ray direction. The x, y and z direction cosines shall be normalized such that $(X \text{ Direction Cosine})^2 + (Y \text{ Direction Cosine})^2 + (Z \text{ Direction Cosine})^2 = 1$ within computational round-off error (i.e., the sum of the squares should be between 0.999 and 1.001).
Example Data:	0.0
Valid Data:	Any real number in the interval [-1 ; 1]
Error Conditions:	A file lacking this data is not a valid IES TM-25-13 ray file. Non-normalized direction cosines shall not be fatal error conditions, as it is anticipated that the software using the ray files will likely check, and thus can re-normalize within their own precision values without any impact on reading speed.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Float

4.8.1.5 Y Direction Cosine

Description:	Shall contain the y direction cosine of the ray direction. The x, y and z direction cosines shall be normalized such that $(X \text{ Direction Cosine})^2 + (Y \text{ Direction Cosine})^2 + (Z \text{ Direction Cosine})^2 = 1$ within computational round-off error (i.e., the sum of the squares should be between 0.999 and 1.001).
Example Data:	0.0
Valid Data:	Any real number in the interval [-1 ; 1]
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file. Non-normalized direction cosines shall not be fatal error conditions, as it is anticipated that the software using the ray files will likely check, and thus can re-normalize within their own precision values without any impact on reading speed.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Float

4.8.1.6 Z Direction Cosine

Description:	Shall contain the z direction cosine of the ray direction. The x, y and z direction cosines shall be normalized such that $(X \text{ Direction Cosine})^2 + (Y \text{ Direction Cosine})^2 + (Z \text{ Direction Cosine})^2 = 1$ within computational round-off error (i.e., the sum of the squares should be between 0.999 and 1.001).
Example Data:	1.0
Valid Data:	Any real number in the interval [-1 ; 1]
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file. Non-normalized direction cosines shall not be fatal error conditions, as it is anticipated that the software using the ray files will likely check, and thus can re-normalize within their own precision values without any impact on reading speed.
Required in File:	Yes
Size in Bytes:	4
Data Type:	Float

4.8.1.7 Radiant Flux, Stokes S0

Description:	Shall contain the radiant flux of the ray specified in watts. This radiant flux is a relative value used to determine relative ray power, and the sum of the radiant flux of all rays is not required to equal the total radiant flux specified in Section 4.7.1.5 . The Radiant Flux is synonymous with the S0 component of the Stokes vector.
Example Data:	100.0
Units:	Radiant watts
Valid Data:	Any positive number
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file if Section 4.7.2.3 is set to Option = 1 or Section 4.7.2.6 is set to Option = 1. If both Radiant and Luminous Flux are present, the Radiant Flux shall take precedence. If the Spectral Data Identifier (Section 4.7.1.9) is set to Option = 1, 2, 3, or 4, the Radiant Flux Flag shall be specified. (See Annex B for further information about the use of spectral data, radiant flux, and luminous flux.)
Required in File:	No
Size in Bytes:	4
Data Type:	Float

4.8.1.8 Wavelength

Description:	Shall contain the wavelength of the ray.
Example Data:	350.0
Units:	Nanometers
Valid Data:	Any positive number
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file if Section 4.7.2.4 is set to Option = 1. (See Annex B for further information about the use of spectral data, radiant flux, and luminous flux.)
Required in File:	No
Size in Bytes:	4
Data Type:	Float

4.8.1.9 Luminous Flux, Y Tristimulus

Description:	Shall contain the luminous flux of the ray specified in lumens. The luminous flux is a relative value used to determine relative ray power, and the sum of the luminous fluxes of all rays is not required to equal the total luminous flux specified in Section 4.7.1.4 .
Example Data:	20.8
Units:	Lumens
Valid Data:	Any positive number, or zero if Radiant Flux (Section 4.8.1.7) is also present.
Error Conditions:	A file lacking this data is not a valid IES TM-25-13 ray file if Section 4.7.2.5 is set to Option = 1. If both Radiant and Luminous Flux are present, the Radiant Flux shall take precedence. If the Spectral Data Identifier (Section 4.7.1.9) is set to Option = 1, 2, 3, or 4, the Luminous Flux shall not be specified for each ray. (See Annex B for further information about the use of spectral data, radiant flux, and luminous flux.)
Required in File:	No
Size in Bytes:	4
Data Type:	Float

4.8.1.10 Stokes S1

Description:	Shall contain the S1 component of the Stokes Vector. The S1, S2 and S3 components of the Stokes vector shall be normalized such that the polarization fraction is equal to $(S1)^2 + (S2)^2 + (S3)^2$.
Example Data:	1.0
Valid Data:	Any number from -1 to 1, including zero
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file if Section 4.7.2.6 is set to Option = 1. Non-normalized Stokes vectors shall be a fatal error.
Required in File:	No
Size in Bytes:	4
Data Type:	Float

4.8.1.11 Stokes S2

Description:	Shall contain the S2 component of the Stokes Vector.
Example Data:	0.0
Valid Data:	Any number from -1 to 1, including zero
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file if Section 4.7.2.6 is set to Option = 1. Non-normalized Stokes vectors shall be a fatal error.
Required in File:	No
Size in Bytes:	4
Data Type:	Float

4.8.1.12 Stokes S3

Description:	Shall contain the S3 component of the Stokes Vector.
Example Data:	0.0
Valid Data:	Any number from -1 to 1, including zero.
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file if Section 4.7.2.6 is set to Option = 1. Non-normalized Stokes vectors shall be a fatal error.
Required in File:	No
Size in Bytes:	4
Data Type:	Float

4.8.1.13 Polarization Ellipse Major Axis X Direction Cosine

Description:	Shall contain the x direction cosine of a vector that is orthogonal to the ray direction unit vector and points in the major axis of the polarization ellipse described by the Stokes vector.
Example Data:	1.0
Valid Data:	Any number from -1 to 1, including zero.
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file if Section 4.7.2.6 is set to Option = 1. Non-normalized Polarization Ellipse vectors shall be a fatal error.
Required in File:	No
Size in Bytes:	4
Data Type:	Float

4.8.1.14 Polarization Ellipse Major Axis Y Direction Cosine

Description:	Shall contain the y direction cosine of a vector that is orthogonal to the ray direction unit vector and points in the major axis of the polarization ellipse described by the Stokes vector.
Example Data:	0.0
Valid Data:	Any number from -1 to 1, including zero.
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file if Section 4.7.2.6 is set to Option = 1. Non-normalized Polarization Ellipse vectors shall be a fatal error.
Required in File:	No
Size in Bytes:	4
Data Type:	Float

4.8.1.15 Polarization Ellipse Major Axis Z Direction Cosine

Description:	Shall contain the z direction cosine of a vector that is orthogonal to the ray direction unit vector and points in the major axis of the polarization ellipse described by the Stokes vector.
Example Data:	0.0
Valid Data:	Any number from -1 to 1, including zero.
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file if Section 4.7.2.6 is set to Option = 1. Non-normalized Polarization Ellipse vectors shall be a fatal error.
Required in File:	No
Size in Bytes:	4
Data Type:	Float

4.8.1.16 X Tristimulus

Description:	Shall contain the X Tristimulus Value. Using full spectral data is preferred over Tristimulus values, which are mainly included for compatibility reasons.
Example Data:	95.047
Valid Data:	Any positive number or zero.
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file if Section 4.7.2.7 is set to Option = 1. (See Annex B for further information about the use of spectral data, radiant flux, and luminous flux.)
Required in File:	No
Size in Bytes:	4
Data Type:	Float

4.8.1.17 Z Tristimulus

Description:	Shall contain the Z Tristimulus Value. Using full spectral data is preferred over Tristimulus values, which are mainly included for compatibility reasons.
Example Data:	108.883
Valid Data:	Any positive number or zero.
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file if Section 4.7.2.7 is set to Option = 1. (See Annex B for further information about the use of spectral data, radiant flux, and luminous flux.)
Required in File:	No
Size in Bytes:	4
Data Type:	Float

4.8.1.18 Spectrum Index

Description:	Shall contain an index to the spectrum to be used for this ray from the spectrum tables in Section 4.7.4 . The first spectrum table shall be given an index of 1, with M as the total number of tables.
Example Data:	1
Valid Data:	Any positive integer less than or equal to Number of Spectral Tables (Section 4.7.1.13).
Error Conditions:	A file lacking this data is not a valid IES-TM-25-13 ray file if Section 4.7.2.8 is set to Option = 1. The spectrum Index shall point to a valid spectrum listed in Section 4.7.4 , or the file is invalid. (See Annex B for further information about the use of spectral data, radiant flux, and luminous flux.)
Required in File:	No
Size in Bytes:	4
Data Type:	Int

4.8.1.19 Additional User Defined Data

Description:	Shall contain all additional user defined data per ray as quantified in Section 4.7.1.14 and named in Section 4.7.5 .
Example Data:	345.8
Valid Data:	Any real number or zero
Error Conditions:	Mismatches between the Number of Additional Ray Data Items Per Ray in Section 4.7.1.14 and the amount of data presented in each ray data line will cause processing errors in the file, resulting in an invalid file.
Required in File:	No
Size in Bytes:	4 bytes per additional data element
Data Type:	Float

5.0 Branding – Logo and Name

The major goal of this document is to create a unified, industry standard format for ray files. Given that there are many existing ray files in the industry, each used only for a specific piece of software, confusion about the new unified format among consumers may arise.

To help eliminate this confusion, ray files conforming to this document shall always be referred to as an “IES TM-25-13 Ray File.” Whenever the file is displayed for download or distribution, it shall be shown with “IES TM-25-13” as a descriptor, and the file extension of “.tm25ray” shall be used.

A logo and icon can be used to identify IES-TM-25-13 ray files. This logo can be applied as an icon for the .tm25ray file type by any software that uses ray files; it can also be used on websites and in publications to show either that the files are IES-TM-25-13 ray files or that the software supports the IES TM-25-13 ray file format. The logo is shown in **Figure 5-1** and the icon in **Figure 5-2**, and both are released under the CC0, “No Rights Reserved” licensing scheme into the Public Domain. Vector logo files can be obtained from ies.org.



Figure 5-1. TM-25 Logo



Figure 5-2. TM-25 Icon

Annex A – File Format Selection

A sub-group of the IES Computer Committee's Subcommittee on Ray File Format for LED Packages and Other Light Sources met to discuss various file format options. The sub-group was tasked with reviewing inputs from the committee to select a format and define the contents of the format according to the agreed data requirements developed by the committee as a whole. Three specific directions were presented to the subcommittee:

- ASCII format
- Use of existing standard formats
- Binary format

A.1 ASCII Format

ASCII was considered as a desirable format, as it is easily readable for all potential users and can easily be used to create and edit files. ASCII files are also very portable over a variety of hardware and software operating systems. Some disadvantages of ASCII are: limits to numerical precision without many characters to represent the data, and the vast file sizes anticipated when files contain millions to billions of rays. Considering the file size as the principle deficiency, the subcommittee opted against using an ASCII format. The software needed to decode binary files to an ASCII representation would likely be available to help users investigate the contents of data files.

A.2 XML Format

XML was considered as a comprehensive standard format. XML is widely used in web design for its extensibility. Extensibility was highly valued by the committee to ensure longevity for the file format. XML is an ASCII format suffering the same deficiencies of a pure ASCII file. Options to use XML for header information combined with binary ray data did not prove to be available in any compact stand form, so XML was also released from consideration.

A.3 HDF5 Format

HDF5 was also considered as a standard format, with advantages of various data types and extensibility. This format was deemed to have too much overhead, requiring the use of APIs (application programming interfaces) that had potential licensing restrictions for some commercial software and metrology vendors.

A.4 Binary Format

A defined binary format was selected as the best choice because of the limited complexity of the data to be stored, a compact size for large data sets, and efficiency in reading and writing the data. The subcommittee considered the trade off in extensibility versus file size and deemed smaller file size to be of predominate value for producers and consumers of ray data. A limited set of extensibility could be offered in a defined format to capture data for known applications.

Annex B – Spectral Data Commentary

To achieve flexibility, the ray file format allows for the transfer of radiant power information in either watts or lumens, and the transfer of color information as tristimulus values, single wavelengths, or entire spectral power distributions. A result of this flexibility is a multitude of cases where data are dependent on other data and/or may not coexist well with other data.

Tristimulus values are mainly included for backward compatibility with existing ray files. Using wavelengths or full spectral power distributions is the preferred method of transferring color information, due to higher fidelity.

The IES TM-25-13 ray file may have the following Spectral Data Identifiers (**Section 4.7.1.9**):

- 0 – No Wavelength Information (watts/lumen based file),
- 1 – Single Wavelength for all rays (wavelength defined in **Section 4.7.1.10**),
- 2 – Wavelength defined per ray (in **Section 4.8.1.8**),
- 3 – Single Spectrum for all rays (spectrum defined in **Section 4.7.4**), or
- 4 – Spectrum defined per ray (spectra defined in **Section 4.7.4**).

Within the ray file, flux values are included in two locations. The first is in the File Header, as **Section 4.7.1.4** and **Section 4.7.1.5**. These sections describe the total overall luminous or radiant flux of the light source, either in lumens or radiant watts, respectively. The second location is in the individual ray data, **Sections 4.8.1.7** and **4.8.1.9**. These sections describe the relative radiant and luminous flux of each ray, respectively, used to determine the distribution of the rays.

The total overall radiant flux of the light source may always be specified in lumens, watts, or both. This situation will allow color models, such as for white LEDs, to be specified in terms of luminous flux, with which users will be more comfortable since LEDs are typically provided this way rather than in radiant flux.

For the flux values assigned to each ray, using lumens or watts, presents a more complicated decision. The ray flux values are used to determine the relative power assigned to each ray. In some cases, the spectral content of each ray varies, and so specifying the relative ray power in lumens or watts may not maintain the correct power ratios. In cases where the color properties are not uniform from ray to ray, the ray flux shall be specified in watts. In cases where the color properties are constant from ray to ray, the ray flux may be specified in lumens.

With Option = 0, 1, or 3, there is no color variation from ray to ray, so the ratio of luminous flux to radiant flux should be constant for each ray. For this reason, the ray flux may be specified in either lumens or watts.

However, there may be ray files in existence that have been converted to the IES TM-25-13 format that specify no color information (Option = 0) and provide both radiant and luminous flux values per ray. These files may not have a constant ratio of luminous flux to radiant flux. This result indicates that there is some color variation over the beam; however, there is no information about what those colors are. For this reason, if software detects both radiant and luminous flux, and a non-constant ratio, it should identify this to the user and let them choose if the distribution should be governed by the luminous or radiant flux values assigned to each ray.

With Option 3, if more than one spectrum is defined in **Section 4.7.4**, the first spectrum shall be used by default. The ray tracing software may offer the functionality to assign any one of the other spectra to the ray set.

With Option = 2 or 4, there is color variation from ray to ray. This means that the ratio of luminous flux to radiant flux may not be constant for each ray. To ensure a proper ray distribution, the ray flux shall be specified in radiant watts and not in luminous flux. To minimize potential errors, the file format requires the use of radiant watts and prohibits the use of lumens for Options 2 and 4.

Annex C – Future Revisions and Compatibility

In proposing any standard file format, there are always features that are wanted by some set of the community but may not be immediately necessary. The IES TM-25-13 file format was designed to serve the basic need of providing a standard ray file format that covers most of the use cases of current ray files that are used in industry.

One goal of the committee was to create a file format such that a parser written for an IES TM-25-13 ray file should be able to read future files and still parse the vast majority of the critical ray data.

To achieve this goal, it is anticipated that the File Header, Known Data Flags, and Description Header sizes will remain fixed for all versions of the IES TM-25-13 ray file. There is extra room in the Fixed Header for future expansion, as well as a mechanism to identify the size of the Additional Text Block that a 2013 parser may simply skip over.

Future revisions will simply insert data in the end of the header, in the Additional Text Block or at the end of each Ray Data Line, which a 2013 parser will know to ignore since the size of this inserted data is known to the 2013 parser.

Alignment of data in chunk sizes that are powers of two is important in order to increase the processing speed of certain computer hardware, and for making software development easier in certain programming environments. To be on the safe side for future developments, the header layout is arranged into blocks whose size is always a multiple of 32 bytes. Data is appended ("padded") with the appropriate number of unused bytes to achieve proper alignment.

Process for Change to an ANSI/IES Standard Under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard Under Continuous Maintenance

NOTE:

Use a separate form for each comment. Submit to the Director of Standards, IES, 85 Broad St, FL 17, New York NY 10004.
Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
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I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

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2. Title of publications and year published _____
3. Clause (section), sub-clause or paragraph number; and page number: _____
4. My proposal (check one):
- ☐ Change to read as follows
 - ☐ Delete and substitute as follows
 - ☐ Add new text as follows
 - ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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ISBN# 978-0-87995-495-6

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